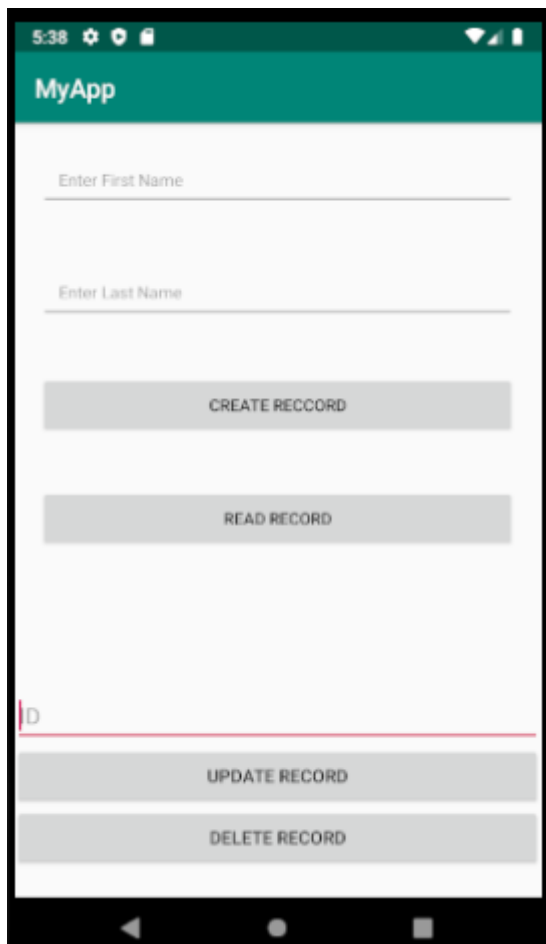


Android CRUD Application



Design SQLite Open Helper class as follow

```
package com.example.myapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;
import android.widget.Toast;

import java.util.ArrayList;
import java.util.List;

public class MySQLiteOpenHelper extends SQLiteOpenHelper {
    public static final String DB_LOG = "DB";
    private static final String DB_NAME = "student.db";
    private static final int DB_VERSION = 1;

    private static final String STUDENT_TABLE_NAME =
"tbl_students";
```

```

    private static final String ID = "id";
    private static final String FIRST_NAME = "first_name";
    private static final String LAST_NAME = "last_name";
    private static final String STUDENT_TABLE_SQL = "CREATE
TABLE " + STUDENT_TABLE_NAME + "(" +
        ID + " INTEGER PRIMARY KEY AUTOINCREMENT," +
        FIRST_NAME + " TEXT, " +
        LAST_NAME + " TEXT)";

    public MySQLiteOpenHelper(Context context){
        super(context, DB_NAME, null, DB_VERSION);

Toast.makeText(context, "Created", Toast.LENGTH_LONG).show();
        Log.i(DB_LOG, "Database Created Successfully");
    }

    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        Log.i(DB_LOG, STUDENT_TABLE_SQL);
        sqLiteDatabase.execSQL(STUDENT_TABLE_SQL);
        Log.i(DB_LOG, "Table Created Successfully");
    }

    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int
oldVersion, int newVersion) {
        String sql = "DROP TABLE IF EXISTS
"+STUDENT_TABLE_NAME;
        sqLiteDatabase.execSQL(sql);
        onCreate(sqLiteDatabase);
    }

    public boolean insert(String firstName, String lastName) {
        ContentValues values = new ContentValues();
        values.put(FIRST_NAME, firstName);
        values.put(LAST_NAME, lastName);

        SQLiteDatabase db = this.getWritableDatabase();
        boolean ins = db.insert(STUDENT_TABLE_NAME, null,
values) > 0;
        Log.i(DB_LOG, "Inserted");
        db.close();
        return ins;
    }

    public List<String> readAll() {
        SQLiteDatabase db = getReadableDatabase();
        String sql = "SELECT * FROM " + STUDENT_TABLE_NAME;
        List<String> allStudents = new ArrayList<>();

        Cursor c = db.rawQuery(sql, null);

```

```

        if (c != null) {
            if (c.moveToFirst()) {
                while(!c.isAfterLast()) {
                    Integer id =
c.getInt(c.getColumnIndex(ID));

                    String student =id+ ". "
+c.getString(c.getColumnIndex(FIRST_NAME)) +"
"+c.getString(c.getColumnIndex(LAST_NAME));
                    allStudents.add(student);
                    c.moveToNext();
                }
            }
        }
        db.close();
        return allStudents;
    }

    public boolean updateRecord(long id,String firstName,
String lastName)
    {
        SQLiteDatabase db = getWritableDatabase();
        ContentValues args = new ContentValues();
        args.put(ID, id);
        args.put(FIRST_NAME, firstName);
        args.put(LAST_NAME, lastName);
        int i = db.update(STUDENT_TABLE_NAME, args, ID + "="
+ id, null);
        return i > 0;
    }

    public boolean deleteRecord(Integer id){
        SQLiteDatabase db = getWritableDatabase();
        // db.execSQL("DELETE FROM " + STUDENT_TABLE_NAME+ "
WHERE "+ID+"=" +id+"");
        return db.delete(STUDENT_TABLE_NAME, ID + "=" + id,
null) > 0;

        //
    }
}

```

Design Layout as Follow:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"

```

```

android:layout_height="match_parent"
tools:context=".CrudActivity"
android:orientation="vertical">

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/et_first_name"
    android:hint="Enter First Name"
    style="@style/TextViewStyle"/>

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/et_last_name"
    android:hint="Enter Last Name"
    style="@style/TextViewStyle"/>

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/createRecord"
    android:text="Create Reccord"
    style="@style/TextViewStyle"/>

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/readRecord"
    android:text="Read Record"
    style="@style/TextViewStyle"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/readRecord"
    android:id="@+id/db_result"
    style="@style/TextViewStyle"/>

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etId" />

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/btnUpdate"
    android:text="Update Record"/>

```

```

        <Button
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:id="@+id/btnDelete"
            android:text="Delete Record"/>

    </LinearLayout>

```

Design Activity as Follow:

```

package com.example.myapplication;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class CrudActivity extends AppCompatActivity {
    private MySQLiteOpenHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_crud);
        db = new MySQLiteOpenHelper(this);

        Button btnCreate = findViewById(R.id.createRecord);
        final Button btnRead = findViewById(R.id.readRecord);
        final Button btnUpdate = findViewById(R.id.btnUpdate);
        final Button btnDelete = findViewById(R.id.btnDelete);

        final EditText etId = findViewById(R.id.etId);
        final EditText etFirstName =
findViewById(R.id.et_first_name);
        final EditText etLastName =
findViewById(R.id.et_last_name);

        btnCreate.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {

                if
(db.insert(etFirstName.getText().toString(),
etLastName.getText().toString())) {
                    Toast.makeText(getApplicationContext(),

```

```

"Record Saved", Toast.LENGTH_LONG).show();
    } else {
        Toast.makeText(getApplicationContext(),
"Unable to Save Record", Toast.LENGTH_LONG).show();
    }
}

});

btnRead.setOnClickListener(new View.OnClickListener()
{
    @Override
    public void onClick(View v) {
        TextView tvDbResult =
findViewById(R.id.db_result);
        StringBuffer buffer = new StringBuffer();
        for (String s : db.readAll()) {
            buffer.append(s + "\n");
        }
        tvDbResult.setText(buffer);
    }
});

btnUpdate.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Integer id =
Integer.parseInt(etId.getText().toString());
        if (db.updateRecord(id,
etFirstName.getText().toString(),
etLastName.getText().toString())) {
            Toast.makeText(getApplicationContext(),
"Record Updated", Toast.LENGTH_LONG).show();
        } else {
            Toast.makeText(getApplicationContext(),
"Sorry unable to Updated", Toast.LENGTH_LONG).show();
        }
    }
});

btnDelete.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Integer id =
Integer.parseInt(etId.getText().toString());
        if (db.deleteRecord(id)) {
            Toast.makeText(getApplicationContext(),
"Record Deleted", Toast.LENGTH_LONG).show();
        } else {

```

```
        Toast.makeText(getApplicationContext(),
"Sorry unable to Delete", Toast.LENGTH_LONG).show();
    }
    });
}
}
```